

Deep Learning-Based Detection of Phishing Websites

Week 3 Report

David Toluwani Ehigie

March 11, 2026

1 Introduction

Week 3 focused on improving the deep learning models developed in Week 2 by integrating engineered features with neural network representations. The goal was to develop a hybrid model that combines the strengths of classical machine learning features and deep learning pattern recognition.

2 Objectives

The main objectives of Week 3 were:

- Identify the most important engineered features
- Perform feature selection
- Develop a hybrid deep learning architecture
- Evaluate the hybrid model against previous models

3 Feature Importance Analysis

A Random Forest classifier was used to compute feature importance scores for all engineered features generated in Week 1.

Based on the importance scores, the top 20 most influential features were selected for the hybrid model.

Feature selection reduces model complexity and improves generalization by removing less informative features.

4 Hybrid Model Architecture

The hybrid model combines two different inputs:

- Character-level URL representation
- Top 20 engineered features

The architecture consists of two branches:

4.1 CNN Branch

The URL input is processed using a Convolutional Neural Network:

- Character embedding layer
- 1D convolution layer
- Global max pooling

This branch captures local character patterns commonly found in phishing URLs.

4.2 Feature Branch

The engineered features are provided as a tabular input directly to the model.

4.3 Feature Fusion

The outputs of both branches are concatenated and passed through dense layers for final classification.

5 Model Training

The model was trained using:

- Binary cross-entropy loss
- Adam optimizer
- Batch size: 64
- Epochs: 10

Dropout regularization was applied to reduce overfitting.

6 Results

The hybrid model achieved the following performance on the test dataset:

Metric	Value
Accuracy	93.2%
Loss	0.220

7 Model Comparison

Model	Input Type	Accuracy
Random Forest	Engineered Features	96.9%
LSTM	URL Only	87.3%
CNN	URL Only	91.9%
Hybrid CNN + Features	Combined Input	93.2%

8 Discussion

The results show that combining deep learning with engineered features improves performance compared to using URL data alone. However, the Random Forest model still achieved the highest accuracy due to its strong performance on structured tabular features.

9 Conclusion

Week 3 successfully demonstrated the effectiveness of hybrid architectures for phishing detection. The hybrid model improved deep learning performance while maintaining strong generalization capability.

Future work may explore more advanced architectures such as attention-based models or transformer networks.